Continuous Assessment 8

Roxy Campher (CMPROX001) and Isabella Lethbridge (LTHISA001)

Question 1

a) Regression coefficients for:

i) Maize yield against soil PH and distance to water.

```
# Extract variables
y_i <- as.matrix(data_tidy_yield$CropYield)

X <- cbind(
  1,
  data_tidy_yield$SoilPH,
  data_tidy_yield$DistanceToWater
)

# Compute beta_hat
XtX <- t(X) %*% X
XtX_inv <- solve(XtX)
XtY <- t(X) %*% y_i

beta_hat <- XtX_inv %*% XtY

beta_table <- data.frame(
  Coefficient = c("Intercept", "SoilpH", "DistanceToWater"),
  Estimate = beta_hat
)

kable(beta_table, digits = 4, caption = "Estimated Regression Coefficients")
```

Table 1: Estimated Regression Coefficients

Coefficient	Estimate
Intercept	20.7219
SoilpH	1.8294
DistanceToWater	-0.0465

ii) Sunlight against soil PH and distance to water.

```
# Extract variables
y_ii <- as.matrix(data_tidy_yield$Sunlight)

X <- cbind(
  1,
  data_tidy_yield$SoilPH,
  data_tidy_yield$DistanceToWater
)

# Compute beta_hat
XtX <- t(X) %*% X
XtX_inv <- solve(XtX)
XtY <- t(X) %*% y_ii

beta_hat <- XtX_inv %*% XtY

beta_table <- data.frame(
  Coefficient = c("Intercept", "SoilpH", "DistanceToWater"),
  Estimate = beta_hat
)

kable(beta_table, digits = 4, caption = "Estimated Regression Coefficients")
```

Table 2: Estimated Regression Coefficients

Coefficient	Estimate
Intercept	10.8590
SoilpH	-0.0089
DistanceToWater	0.0445

iii) Maize yield and sunlight against soil PH and distance to water.

```r
# MV Regression
Y_iii <- as.matrix(data_tidy_yield[, c("CropYield", "Sunlight")])

# Design matrix
X <- cbind(
  1,
  data_tidy_yield$SoilPH,
  data_tidy_yield$DistanceToWater
)

# OLS estimator
XtX <- t(X) %*% X
XtX_inv <- solve(XtX)
XtY <- t(X) %*% Y_iii

B_hat <- XtX_inv %*% XtY

beta_table <- data.frame(
  Predictor = c("Intercept", "SoilpH", "DistanceToWater"),
  MaizeYield = B_hat[, 1],
  Sunlight = B_hat[, 2]
)

kable(beta_table, digits = 4, caption = "Estimated Coefficients for Multivariate Regression")
```

Table 3: Estimated Coefficients for Multivariate Regression

Predictor	MaizeYield	Sunlight
Intercept	20.7219	10.8590
SoilpH	1.8294	-0.0089
DistanceToWater	-0.0465	0.0445

b) Confidence interval for the effect of soil PH on temperature:

```r
# Extract variables
y <- as.matrix(data_tidy_yield$TempC)
x <- as.matrix(data_tidy_yield$SoilPH)

# Design matrix
X <- cbind(1, x)
```

```r
# Beta estimate
XtX <- t(X) %*% X
XtX_inv <- solve(XtX)
XtY <- t(X) %*% y
beta_hat <- XtX_inv %*% XtY

# Residuals
y_hat <- X %*% beta_hat
residuals <- y - y_hat

n <- nrow(X)
k <- ncol(X)

sigma2_hat <- as.numeric(t(residuals) %*% residuals / (n - k))

# Standard error of SoilpH coefficient
var_beta <- sigma2_hat * XtX_inv
se_beta1 <- sqrt(var_beta[2, 2])

# t critical value
t_crit <- qt(0.975, df = n - k)

# Confidence interval
beta1 <- beta_hat[2]

CI <- c(
  beta1 - t_crit * se_beta1,
  beta1 + t_crit * se_beta1
)

ci_table <- data.frame(
  Term = "SoilpH effect on Temp",
  Lower_95 = CI[1],
  Upper_95 = CI[2]
)

kable(ci_table, digits = 4)
```

Term	Lower_95	Upper_95
SoilpH effect on Temp	0.791	2.3183

c) F-test for the effect of soil PH and distance to water on maize yield.

```
# Data
y <- as.matrix(data_tidy_yield$CropYield)

X <- cbind(
  1,
  data_tidy_yield$SoilPH,
  data_tidy_yield$DistanceToWater
)

n <- nrow(X)
k <- ncol(X)

# OLS estimates
XtX <- t(X) %*% X
C <- solve(XtX)
beta_hat <- C %*% t(X) %*% y

# Residual variance
res <- y - X %*% beta_hat
s2 <- as.numeric(t(res) %*% res / (n - k))

# Extract Beta(1): SoilpH + DistanceToWater
beta1_hat <- beta_hat[2:3, , drop = FALSE]

# Extract C11 block
C11 <- C[2:3, 2:3]

# F statistic
q <- 2

F_stat <- as.numeric(
  t(beta1_hat) %*% solve(C11) %*% beta1_hat / (q * s2)
)

p_value <- 1 - pf(F_stat, df1 = 2, df2 = n - k)

p_statss <- data.frame(
  Term = "F-test results:",
  F_Statistic = round(F_stat, 3),
  P_Value = p_value
)
```

```
p_statss
```

```
              Term F_Statistic      P_Value
1 F-test results:      29.103 4.634138e-11
```

The null hypothesis is $H_0 : \beta_{pH} = \beta_{distance} = 0$ against the alternative that at least one coefficient is non-zero.

The p-value is very small so we can reject H_0 with confidence and conclude that soil PH and distance to water jointly have an effect on maize yield.

d) Likelihood ratio test for the effect of soil PH on either maize yield or sunlight.

```
Y <- as.matrix(data_tidy_yield[, c("CropYield", "Sunlight")])

X_full <- cbind(
  1,
  data_tidy_yield$SoilPH,
  data_tidy_yield$Rainfall,
  data_tidy_yield$TempC,
  data_tidy_yield$WindSpeed,
  data_tidy_yield$DistanceToWater,
  data_tidy_yield$Altitude
)

X_res <- cbind(
  1,
  data_tidy_yield$Rainfall,
  data_tidy_yield$TempC,
  data_tidy_yield$WindSpeed,
  data_tidy_yield$DistanceToWater,
  data_tidy_yield$Altitude
)

B_mat_full <- solve(t(X_full) %*% X_full) %*% t(X_full) %*% Y
B_mat_res  <- solve(t(X_res) %*% X_res) %*% t(X_res) %*% Y

Sigma_mat_full <- t((Y - X_full %*% B_mat_full)) %*% (Y - X_full %*% B_mat_full) / (nrow(Y))

Sigma_mat_res <- t((Y - X_res %*% B_mat_res)) %*% (Y - X_res %*% B_mat_res) / (nrow(Y))

n <- nrow(Y)
```

```r
r <- ncol(Y)
q <- ncol(X_full) - ncol(X_res)
k <- ncol(X_full)
p <- k - 1

test_stat <- -(n - k - 0.5 * (r - p + q + 1)) * log(det(Sigma_mat_full) / det(Sigma_mat_res)

#test_stat |> signif(2)

df <- r * q

p_value <- pchisq(test_stat, df = df, lower.tail = FALSE)

lrt_results <- data.frame(
  Factor = "SoilPH",
  LRT_Statistic = test_stat,
  DF = df,
  P_Value = p_value
)

lrt_results
```

```
  Factor LRT_Statistic DF      P_Value
1 SoilPH      48.81475  2 2.511949e-11
```

The null hypothesis states that $H_0 : \beta_{soilPH,yield} = 0$ and $\beta_{soilPH,sunlight} = 0$, the null hypothesis is that soil PH has no effect on Maize yield and sunlight.

The p-value is very small so we can confidently reject H_0 and conclude that soil pH has a significant effect on at least one of the response variables (crop yield or sunlight).

Question 2

```r
# Data
Y <- as.matrix(data_tidy_mali$cotton)

X <- data_tidy_mali[, names(data_tidy_mali) != "cotton"]
X <- as.matrix(X)

# Scale vars
```

```
X_scaled <- scale(X)
Y_centered <- scale(Y, center = TRUE, scale = FALSE)
```

Multiple regression:

```
beta_ols <- solve(t(X_scaled) %*% X_scaled) %*% t(X_scaled) %*% Y_centered

y_hat_ols <- X_scaled %*% beta_ols

mse_ols <- mean((Y_centered - y_hat_ols)^2)
```

PCR:

```
S <- t(X_scaled) %*% X_scaled / nrow(X_scaled)

eig <- eigen(S)

V <- eig$vectors   # loadings

Z <- X_scaled %*% V

Z <- X_scaled %*% V # PC's

k <- 6 # first k components
Z_k <- Z[, 1:k]

theta_pcr <- solve(t(Z_k) %*% Z_k) %*% t(Z_k) %*% Y_centered # OLS in PC space

# Predictions
y_hat_pcr <- Z_k %*% theta_pcr
mse_pcr <- mean((Y_centered - y_hat_pcr)^2)
```

PLS:

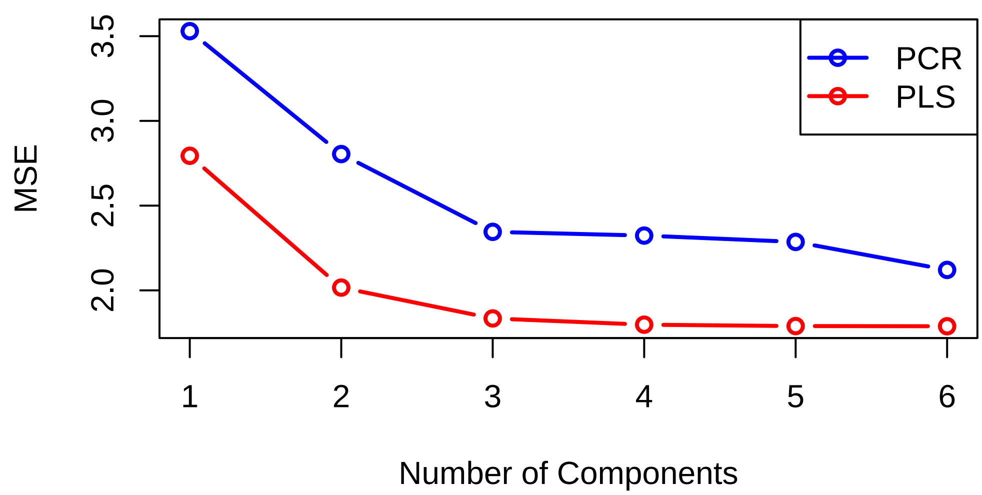
```
n <- nrow(X_scaled)
p <- ncol(X_scaled)

X_m <- X_scaled
y_m <- Y_centered

y_hat_pls <- rep(0, n)
```

```r
M <- 4   # tune this

for (m in 1:M) {

  # (a) direction weights
  phi <- t(X_m) %*% y_m

  z <- X_m %*% phi   # latent component

  # (b) coefficient
  theta <- sum(z * y_m) / sum(z^2)

  # (c) update fitted values
  y_hat_pls <- y_hat_pls + theta * z

  # (d) deflate X
  for (j in 1:p) {
    X_m[, j] <- X_m[, j] - (sum(z * X_m[, j]) / sum(z^2)) * z
  }
}

mse_pls <- mean((Y_centered - y_hat_pls)^2)
```

Number of components:

```r
max_comp <- 6
mse_pcr_vec <- rep(0, max_comp)
mse_pls_vec <- rep(0, max_comp)

for (k in 1:max_comp) {

  # PCR
  Z_k <- Z[, 1:k]
  theta <- solve(t(Z_k) %*% Z_k) %*% t(Z_k) %*% Y_centered # OLS reg in PC space
  mse_pcr_vec[k] <- mean((Y_centered - Z_k %*% theta)^2) # pred error

  # PLS
  X_m <- X_scaled
  y_hat <- rep(0, n)

  for (m in 1:k) { # Build PLS components sequentially
```

```
    phi <- t(X_m) %*% y_m
    z <- X_m %*% phi
    theta <- sum(z * y_m) / sum(z^2)
    y_hat <- y_hat + theta * z

    for (j in 1:p) {
      X_m[, j] <- X_m[, j] - (sum(z * X_m[, j]) / sum(z^2)) * z
    }
  }

  mse_pls_vec[k] <- mean((Y_centered - y_hat)^2)
}
```



As shown in the plot MSE decreases for both methods as the number of components increase, which is expected as more variation is being captured. From the plot we see that PLS requires less components before it reaches a stable error level compared to PCR. Based on the MSE curves, 4 components were selected for PCR and 6 components were selected for PLS as reasonable choices. The corresponding MSE values are summarized below:

Table 5: MSE Comparison across methods.

OLS_MSE	PCR_MSE	PLS_MSE
1.7874	2.1206	1.7974

OLS performs best overall, achieving the lowest MSE, indicating that using all predictors without dimensionality reduction already provides strong predictive accuracy for cotton production.

PCR performs worst, with a higher MSE than OLS. This is expected because PCR constructs components based on variance in the predictors only, leading to components that are not necessarily predictive of the outcome. PLS performs close to OLS and substantially better than PCR. This is because PLS constructs components that maximise covariance between predictors and the response, making them more relevant for prediction.

Overall, PLS provides the best trade-off among the reduced-dimension methods, while PCR is less effective in this setting. However, since OLS performs slightly better than both, dimensionality reduction does not significantly improve predictive accuracy for this dataset.